

Exercise 0.1 Let V and W be discrete random variables defined on some probability space with a joint pmf $P_{V,W}(v, w)$. We do not assume independence.

- a.) Prove that $\mathbb{E}[V + W] = \mathbb{E}[V] + \mathbb{E}[W]$.
- b.) Prove that if V and W are independent, then $\mathbb{E}[VW] = \mathbb{E}[V]\mathbb{E}[W]$.
- c.) Let V and W be independent and let σ_V^2 and σ_W^2 be their respective variances. Find the variance of $Z = V + W$.

Solution:

- a.) We compute $\mathbb{E}[V + W]$. By definition

$$\begin{aligned}\mathbb{E}[V + W] &= \sum_{v,w} (v + w) p_{V,W}(v, w) = \sum_{v,w} v p_{V,W}(v, w) + \sum_w w p_{V,W}(v, w) \\ &= \sum_v v p_V(v) + \sum_w w p_W(w) = \mathbb{E}[V] + \mathbb{E}[W].\end{aligned}$$

- b.) We compute $\mathbb{E}[VW]$. By definition

$$\begin{aligned}\mathbb{E}[VW] &= \sum_{v,w} v w p_{V,W}(v, w) = \sum_{v,w} v w p_V(v) p_W(w) \\ &= \sum_v v p_V(v) \sum_w w p_W(w) = \mathbb{E}[V]\mathbb{E}[W].\end{aligned}$$

- c.) We compute σ_Z^2 . By definition

$$\begin{aligned}\sigma_Z^2 &= \mathbb{E}[Z^2] - \mathbb{E}[Z]^2 = \mathbb{E}[(V + W)^2] - \mathbb{E}[V + W]^2 \\ &= \mathbb{E}[V^2 + W^2 + 2VW] - (\mathbb{E}[V]^2 + 2\mathbb{E}[V]\mathbb{E}[W] + \mathbb{E}[W]^2) \\ &= \mathbb{E}[V^2] - \mathbb{E}[V]^2 + \mathbb{E}[W^2] - \mathbb{E}[W]^2 = \sigma_V^2 + \sigma_W^2,\end{aligned}$$

where the second to last equality follows from the fact that $\mathbb{E}[VW] = \mathbb{E}[V]\mathbb{E}[W]$.

Exercise 0.2 For a nonnegative integer-valued random variable N , show that $\mathbb{E}[N] = \sum_{n>0} P[N \geq n]$ and that $P[N \geq n] = \mathbb{E}[\mathbf{1}\{N \geq n\}]$.

Solution: We calculate directly:

$$\begin{aligned}\mathbb{E}[N] &= \sum_{n=0}^{\infty} n p_N(n) = 0 \cdot p_N(0) + 1 \cdot p_N(1) + 2 \cdot p_N(2) + 3 \cdot p_N(3) + \dots \\ &= [p_N(1) + p_N(2) + p_N(3) + \dots] \\ &\quad + [p_N(2) + p_N(3) + p_N(4) + \dots] \\ &\quad + [p_N(3) + p_N(4) + p_N(5) + \dots] + \dots \\ &= P[N \geq 1] + P[N \geq 2] + P[N \geq 3] + \dots = \sum_{n>0} P[N \geq n].\end{aligned}$$

The second statement holds because $\mathbb{E}[\mathbf{1}\{N \geq n\}] = 1 \cdot P[N \geq n] + 0 \cdot P[N < n] = P[N \geq n]$.

Exercise 0.3 Flip a fair coin four times. Let X be the number of Heads obtained, and let Y be the position of the first Heads i.e. if the sequence of coin flips is $TTHT$, then $Y = 3$, if it is $THHH$, then $Y = 2$. If there are no heads in the four tosses, then we define $Y = 0$.

- Model the experiment completely, i.e. define the sample space and the random variables X and Y as functions from that sample space.
- Find the joint pmf of X and Y .
- Using the joint pmf, find the marginal pmf of X . What is $P[Y = 0|X = 1]$ and $P[Y = 1|X = 3]$?

Solution:

- The underlying sample space is the set

$$\Omega = \{TTTT, TTTH, \dots, HHHH\}.$$

Each outcome $\omega \in \Omega$ can be mapped to $X(\omega)$ and $Y(\omega)$, e.g.,

$$\begin{aligned} X(TTTT) &= 0 & Y(TTTT) &= 0 \\ X(THHT) &= 2 & Y(THHT) &= 2 \\ X(TTTH) &= 1 & Y(TTTH) &= 4 \\ &\text{etc.} \end{aligned}$$

- By listing all 16 elements of Ω , and computing X and Y for each we can see that, e.g.,

$$\{X = 2, Y = 2\} = \{THHT, THTH\},$$

and thus $P_{XY}(2, 2) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}$. This can be repeated for all feasible values of X and Y as in the table below:

	x				
y	0	1	2	3	4
0	$\frac{1}{16}$				
1		$\frac{1}{16}$	$\frac{3}{16}$	$\frac{3}{16}$	$\frac{1}{16}$
2		$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{16}$	
3		$\frac{1}{16}$	$\frac{1}{16}$		
4		$\frac{1}{16}$			

We can now read off.

- Summing over the columns of the table, we get the pmf of X as

$$P_X(x) = \begin{cases} \frac{1}{16} & x = 0 \\ \frac{1}{4} & x = 1 \\ \frac{3}{8} & x = 2 \\ \frac{1}{4} & x = 3 \\ \frac{1}{16} & x = 4 \end{cases}$$

We can now use Bayes' rule to compute

$$\begin{aligned} P[Y = 0|X = 1] &= \frac{P_{XY}(1, 0)}{P_X(1)} = 0, \\ P[Y = 1|X = 3] &= \frac{P_{XY}(3, 1)}{P_X(3)} = \frac{3}{16} \cdot \frac{4}{1} = \frac{3}{4}. \end{aligned}$$

Exercise 0.4 Derive the addition and multiplication tables for F_8 and F_9 . You should use the construction described in the lecture notes and the irreducible polynomials $x^3 + x + 1$ for F_8 and $x^2 + 1$ for F_9 .

Hint: You may want to use MATLAB to solve this problem. However, you will need to compute some elements by hand to verify the computer-generated output.

Solution:

$+_{F_8}$	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	0	3	2	5	4	7	6
2	2	3	0	1	6	7	4	5
3	3	2	1	0	7	6	5	4
4	4	5	6	7	0	1	2	3
5	5	4	7	6	1	0	3	2
6	6	7	4	5	2	3	0	1
7	7	6	5	4	3	2	1	0

$+_{F_9}$	0	1	2	3	4	5	6	7	8
0	0	1	2	3	4	5	6	7	8
1	1	2	0	4	5	3	7	8	6
2	2	0	1	5	3	4	8	6	7
3	3	4	5	6	7	8	0	1	2
4	4	5	3	7	8	6	1	2	0
5	5	3	4	8	6	7	2	0	1
6	6	7	8	0	1	2	3	4	5
7	7	8	6	1	2	0	4	5	3
8	8	6	7	2	0	1	5	3	4

$\times_{F_8}$	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	3	1	7	5
3	0	3	6	5	7	4	1	2
4	0	4	3	7	6	2	5	1
5	0	5	1	4	2	7	3	6
6	0	6	7	1	5	3	2	4
7	0	7	5	2	1	6	4	3

$\times_{F_9}$	0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8
2	0	2	1	6	8	7	3	5	4
3	0	3	6	2	5	8	1	4	7
4	0	4	8	5	6	1	7	2	3
5	0	5	7	8	1	3	4	6	2
6	0	6	3	1	7	4	2	8	5
7	0	7	5	4	2	6	8	3	1
8	0	8	4	7	3	2	5	1	6

The MATLAB Program generating above tables:

Listing 1: addition_table.m

```

1 p = 3;
2 degree = 1; % Thus the size of the field is p^(degree+1)
3 h = [1,0,1]; % The irreducible polynomial of degree 'degree+1'
4 % List all polynomials
5 F = zeros(p^(degree+1),degree+1);
6 for i = 1:p^(degree+1)
7     m = i-1;
8     for j = 1:degree+1
9         F(i,j) = mod(m,p);
10        m = (m-mod(m,p))/p;
11    end
12 end
13 a_table = zeros(p^(degree+1));
14 for i = 1:p^(degree+1)
15     for j = 1:p^(degree+1)
16         f = mod(F(i,:)+F(j,:),p);
17         [~,k] = ismember(f,F,'rows');
18         a_table(i,j) = k-1;
19     end
20 end
21 disp(a_table);

```

Listing 2: multiplication_table.m

```

1 p = 3;
2 degree = 1; % Thus the size of the field is p^(degree+1)
3 h = [1,0,1]; % The irriducible polynomial of degree 'degree+1'
4 % List all polynomials
5 F = zeros(p^(degree+1),degree+1);
6 for i = 1:p^(degree+1)
7     m = i-1;
8     for j = 1:degree+1
9         F(i,j) = mod(m,p);
10        m = (m-mod(m,p))/p;
11    end
12 end
13 m_table = zeros(p^(degree+1));
14 for i = 1:p^(degree+1)
15     for j = 1:p^(degree+1)
16         f = polyMod(polyMultiply(F(i,:),F(j,:),p),h,p);
17         [~,k] = ismember(f,F,'rows');
18         m_table(i,j) = k-1;
19     end
20 end
21 disp(m_table);

```

Listing 3: polyMultiply.m

```

1 function [ f ] = polyMultiply (g, h, p)
2     % Multiply two polynomials on base field Fp where p is a prime number.
3     g_degree = length(g)-1;
4     h_degree = length(h)-1;
5     f = zeros(1, g_degree + h_degree+1);
6     for k = 0:(length(f)-1) % k-th order
7         for a = max(0,k-h_degree):min(k,g_degree)
8             b = k-a;
9             f(k+1) = f(k+1) + g(a+1)*h(b+1);
10        end
11    end
12    f = mod(f, p);
13 end

```

Listing 4: polyMod.m

```

1 function [ f ] = polyMod (g, h, p)
2     % Find the modulo of polynomial g with respect to irriducible polynomial h on
3     % base field Fp where p is a prime number.
4     g_length = length(g);
5     h_length = length(h);
6     while (max([g ~= 0].*[1:g_length]) >= h_length)
7         d = max([g ~= 0].*[1:g_length]);
8         C = g(d);
9         h_shifted = zeros(size(g));
10        h_shifted(d-h_length+1:d) = h;
11        g = mod(g-C*h_shifted, p);
12    end

```

```

12   f = g(1: h_length-1);
13   end

```

Exercise 0.5 For discrete random variables A and B , derive the Cauchy-Schwarz inequality in the form

$$\mathbb{E}[AB] \leq \sqrt{\mathbb{E}[A^2]\mathbb{E}[B^2]},$$

either from its vector form in the lecture notes, or by calculating the expectation of the non-negative random variable $(X - \alpha Y)^2$, for a suitable choice of α .

Solution: Solution 1, starting from $(X - \alpha Y)^2$: We have

$$0 \leq \mathbb{E}[(X - \alpha Y)^2] = \mathbb{E}[X^2] - 2\alpha\mathbb{E}[XY] + \alpha^2\mathbb{E}[Y^2].$$

Choosing $\alpha = \mathbb{E}[XY]/\mathbb{E}[Y^2]$ yields

$$0 \leq \mathbb{E}[X^2] - 2\frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]} + \frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]},$$

which equals the desired statement after multiplication with $\mathbb{E}[Y^2]$ on both sides.

Solution 2: Let $u_{ij} = x_i\sqrt{P[X = x_i, Y = y_j]}$ and $v_{ij} = y_j\sqrt{P[X = x_i, Y = y_j]}$. The inner product between u and v in the Euclidean space is

$$\begin{aligned}\langle u, v \rangle &= \sum_{ij} x_i y_j P[X = x_i, Y = y_j] = \mathbb{E}[XY], \\ \langle u, u \rangle &= \sum_{ij} x_i^2 P[X = x_i, Y = y_j] = \mathbb{E}[X^2], \\ \langle v, v \rangle &= \sum_{ij} y_j^2 P[X = x_i, Y = y_j] = \mathbb{E}[Y^2].\end{aligned}$$

By the Cauchy-Schwarz inequality for the vectors, we obtain

$$|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle},$$

which is equivalent to

$$\mathbb{E}[XY] \leq \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}.$$

A remark about the vector form of the Cauchy-Schwarz inequality: In order to use the Cauchy-Schwarz inequality, you should argue that $\langle u, v \rangle$ is an inner product. Here we use the vector inner product on the Euclidean space for it. Some people may want to define the inner product as $\langle u, v \rangle = \sum_{ij} u_{ij} v_{ij} P[X = x_i, Y = y_j]$. It is indeed an inner product, but you have to prove it.

Exercise 0.6 (Tail bounds) In this exercise we will derive some further tail bounds.

- a) Derive the one-sided Cheybshev inequality, which says that $P[Y \geq a] \leq \sigma_Y^2/(\sigma_Y^2 + a^2)$ if $\mathbb{E}[Y] = 0$ and $a > 0$.

Hint: You may first show that $a \leq \mathbb{E}[(a - Y)\mathbf{I}\{Y < a\}]$. After that, you may apply the Cauchy-Schwarz inequality on two random variables, $a - Y$ and $\mathbf{I}\{Y < a\}$.

- b) Derive the reverse Markov inequality: Let X be a random variable such that $P[X \leq a] = 1$ for some constant a . Then for $d < \mathbb{E}[X]$, we have

$$P[X > d] \geq \frac{\mathbb{E}[X] - d}{a - d}.$$

Hint: You may apply the Markov inequality on the random variable $\tilde{X} = a - X$.

Solution:

a.) Since Y has zero mean, we have

$$a = \mathbb{E}[a - Y].$$

Considering the expectation above, we have

$$\mathbb{E}[a - Y] = \sum_y P_Y(y)(a - y) = \sum_{y:y < a} P_Y(y)(a - y) + \sum_{y:y \geq a} P_Y(y)(a - y) \leq \sum_{y:y < a} P_Y(y)(a - y),$$

because the second sum is non-positive. This can be written as

$$a = \mathbb{E}[a - Y] \leq \sum_y P_Y(y)(a - y)\mathbf{1}\{y < a\},$$

where $\mathbf{1}\{\text{statement}\}$ returns 1 if the statement is true and 0 otherwise. Thus, we have

$$a \leq \mathbb{E}[(a - Y)\mathbf{1}\{Y < a\}].$$

Now square both sides,

$$a^2 \leq (\mathbb{E}[(a - Y)\mathbf{1}\{Y < a\}])^2.$$

Apply the Cauchy-Schwarz inequality to the expectation,

$$a^2 \leq \mathbb{E}[(a - Y)^2]\mathbb{E}[\mathbf{1}\{Y < a\}^2] = \mathbb{E}[(a - Y)^2]\mathbb{E}[\mathbf{1}\{Y < a\}] = (a^2 + \mathbb{E}[Y^2])P(Y < a).$$

Since $\mathbb{E}[Y^2] = \sigma_Y^2$, rearrangement of the above inequality yields the one-sided Chebyshev inequality as desired.

b.) Apply Markov's inequality to the non-negative random variable $\tilde{X} := a - X$. Then one has

$$P(X \leq d) = P(a - \tilde{X} \leq d) = P(\tilde{X} \geq a - d) \leq \frac{\mathbb{E}[\tilde{X}]}{a - d} = \frac{\mathbb{E}[a - X]}{a - d}.$$

Hence,

$$P(X < d) \geq 1 - \frac{\mathbb{E}[a - X]}{a - d} = \frac{\mathbb{E}[X] - d}{a - d}.$$

Exercise 0.7 (Chernoff bound) Let $X_1, \dots, X_n$ be a sequence of i.i.d. random variables with zero-mean and moment generating function $M_X(s) := \mathbb{E}[e^{sX}]$. Show that for any $\epsilon > 0$,

$$P\left(\frac{1}{n}(X_1 + \dots + X_n) > \epsilon\right) \leq \exp\left(-n \max_{s \geq 0}(\epsilon s - \log M_X(s))\right).$$

Hint: Note that the event $\{\frac{1}{n}(X_1 + \dots + X_n) > \epsilon\}$ occurs exactly if $\{\exp(s(X_1 + \dots + X_n)) > \exp(n\epsilon s)\}$, for any fixed $s \geq 0$. Now apply Markov's inequality.

Solution: Using the hint, we know that

$$\begin{aligned} P\left(\frac{1}{n}(X_1 + \dots + X_n) > \epsilon\right) &= P(\exp(s(X_1 + \dots + X_n)) > \exp(n\epsilon s)) \\ &\leq \frac{\mathbb{E}[\exp(s(X_1 + \dots + X_n))]}{\exp(n\epsilon s)}, \end{aligned}$$

where the inequality is from Markov's inequality. Since the X_i 's are independent,

$$P\left(\frac{1}{n}(X_1 + \dots + X_n) > \epsilon\right) \leq \frac{\prod_{i=1}^n \mathbb{E}[\exp(sX_i)]}{\exp(n\epsilon s)} = \exp[-n(\epsilon s - \log M_X(s))],$$

where the final equality is due to the fact that the X_i are identically distributed. Since $s \geq 0$ is arbitrary, we can minimize over this parameter.

Exercise 0.8 Let f be convex on $[a, b]$. Using only the defining property of convex functions, show that for any $a \leq x_1 < x_2 \leq x_3 < x_4 \leq b$, we have

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} \leq \frac{f(x_4) - f(x_3)}{x_4 - x_3}.$$

Solution: Recall that a function $f : (a, b) \rightarrow \mathbb{R}$ is convex on (a, b) if for all $x, y \in (a, b)$ with $x < y$ and $0 \leq \lambda \leq 1$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y). \quad (1)$$

For brevity in notation, we define $g(x, y) = (f(y) - f(x))/(y - x)$, so we must show that $g(x_1, x_2) \leq g(x_3, x_4)$. We first show that for $x_1, x_2, x_3 \in (a, b)$ such that $x_1 < x_2 < x_3$, we have $g(x_1, x_2) \leq g(x_2, x_3)$, i.e., that

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} \leq \frac{f(x_3) - f(x_2)}{x_3 - x_2}. \quad (2)$$

In (1), we let $x \leftarrow x_1$, $y \leftarrow x_3$ and $\lambda = (x_3 - x_2)/(x_3 - x_1) \in [0, 1]$. Then we check that

$$\lambda x + (1 - \lambda)y = \lambda x_1 + (1 - \lambda)x_3 = \frac{x_3 - x_2}{x_3 - x_1} \cdot x_1 + \frac{x_2 - x_1}{x_3 - x_1} \cdot x_3 = x_2.$$

So we have

$$f(\lambda x + (1 - \lambda)y) = f(x_2) \leq \frac{x_3 - x_2}{x_3 - x_1} f(x_1) + \frac{x_2 - x_1}{x_3 - x_1} f(x_3) = \lambda f(x_1) + (1 - \lambda)f(x_3).$$

This is equivalent to

$$\lambda f(x_2) + (1 - \lambda)f(x_2) = \lambda f(x_1) + (1 - \lambda)f(x_3),$$

or

$$\lambda(f(x_2) - f(x_1)) \leq (1 - \lambda)(f(x_3) - f(x_2))$$

Recalling the definition of λ shows (2), i.e., that $g(x_1, x_2) \leq g(x_2, x_3)$. By the same logic, for three points $x_2 < x_3 < x_4$, we also have $g(x_2, x_3) \leq g(x_3, x_4)$. Combining these two inequalities yields $g(x_1, x_2) \leq g(x_3, x_4)$ as desired.

Exercise 0.9 In this exercise, we aim to get an intuition for the Minimax Theorem. Recall that the theorem states that given two compact convex sets $\mathcal{X}$ and $\mathcal{Y}$, and a continuous convex-concave function $f : (\mathcal{X}, \mathcal{Y}) \rightarrow \mathbb{R}$, it holds that

$$\min_{x \in \mathcal{X}} \max_{y \in \mathcal{Y}} f(x, y) = \max_{y \in \mathcal{Y}} \min_{x \in \mathcal{X}} f(x, y). \quad (3)$$

For this exercise, we fix the function to be

$$f(x, y) = x^2 + 4xy - y^2.$$

- Show that f satisfies the requirements of the theorem and compute $g_{\mathcal{Y}}(x) := \max_{y \in \mathcal{Y}} f(x, y)$ and $h_{\mathcal{X}}(y) := \min_{x \in \mathcal{X}} f(x, y)$ for general intervals $\mathcal{X}$ and $\mathcal{Y}$, that is $\mathcal{X} = [x_{\min}, x_{\max}]$ and $\mathcal{Y} = [y_{\min}, y_{\max}]$ for variables $x_{\min} \leq x_{\max}$ and $y_{\min} \leq y_{\max}$. This requires a case distinction.
- Convince yourself that the theorem holds for the specific choice $\mathcal{X} = [-2, 2]$ and $\mathcal{Y} = [-1, 1]$ by computing both sides of (3) starting from the result in a). Note that one side is more direct than the other.
- Next, we modify $\mathcal{X}$ such that the result of the optimization changes. Let $\mathcal{X} = [x_{\min}, 2]$, where $2 > x_{\min} > 0$, and $\mathcal{Y} = [-1, 1]$. Calculate both sides of (3) as a function of $x_{\min}$.

Solution:

- a.) We first note that $\partial_x^2 f(x, y) = 2 > 0$ and $\partial_y^2 f(x, y) = -2 < 0$. The global extrema of f with respect to x and y are found by setting the first derivative to zero. However, these points may not lie within $\mathcal{X}$ and $\mathcal{Y}$, which necessitates a case distinction as we then have to pick the closest points in the respective interval. Note that

$$\arg \min_{x \in \mathbb{R}} f(x, y) = -2y, \quad \arg \max_{y \in \mathbb{R}} f(x, y) = 2x.$$

We find

$$h_{\mathcal{X}}(y) = \min_{x \in \mathcal{X}} f(x, y) = \begin{cases} x_{\min}^2 + 4x_{\min}y - y^2 & \text{if } x_{\min} > -2y, \\ x_{\max}^2 + 4x_{\max}y - y^2 & \text{if } x_{\max} < -2y, \\ 4y^2 - 8y^2 - y^2 = -5y^2 & \text{otherwise,} \end{cases}$$

$$g_{\mathcal{Y}}(x) = \max_{y \in \mathcal{Y}} f(x, y) = \begin{cases} x^2 + 4xy_{\min} - y_{\min}^2 & \text{if } y_{\min} > 2x, \\ x^2 + 4xy_{\max} - y_{\max}^2 & \text{if } y_{\max} < 2x, \\ x^2 + 8x^2 - 4x^2 = 5x^2 & \text{otherwise.} \end{cases}$$

(Equivalently, one might replace the strict inequalities by $\geq$ and $\leq$ instead.)

- b.) Let $\mathcal{X} = [-2, 2]$ and $\mathcal{Y} = [-1, 1]$. Then $h_{\mathcal{X}}(y)$ simplifies to $h_{\mathcal{X}}(y) = -5y^2$, such that $\max_{y \in \mathcal{Y}} \min_{x \in \mathcal{X}} f(x, y) = 0$. However, $g_{\mathcal{Y}}(x)$ does not simplify as nicely:

$$g_{\mathcal{Y}}(x) = \begin{cases} x^2 - 4x - 1 & \text{if } -1/2 > x, \\ x^2 + 4x - 1 & \text{if } 1/2 < x, \\ 5x^2 & \text{otherwise.} \end{cases}$$

we find the minima individually, the result is again zero:

$$\min_{x \in \mathcal{X}} \max_{y \in \mathcal{Y}} f(x, y) = \min \left\{ \min_{x \in [-2, -1/2]} x^2 - 4x - 1, \min_{x \in [-1/2, 1/2]} 5x^2, \min_{x \in [1/2, 2]} x^2 + 4x - 1 \right\}$$

$$= \min\{0, 5/4\} = 0.$$

- c.) The function $h_{\mathcal{X}}(y)$ simplifies to

$$h_{\mathcal{X}}(y) = \begin{cases} x_{\min}^2 + 4x_{\min}y - y^2 & \text{if } x_{\min} > -2y, \\ 4y^2 - 8y^2 - y^2 = -5y^2 & \text{otherwise,} \end{cases}$$

such that

$$\max_{y \in \mathcal{Y}} \min_{x \in \mathcal{X}} f(x, y) = \max \left\{ \max_{y \in [-1, -x_{\min}/2]} -5y^2, \max_{y \in [-x_{\min}/2, 1]} x_{\min}^2 + 4x_{\min}y - y^2 \right\}$$

$$= \max\{-5x_{\min}^2/4, x_{\min}^2 + 4x_{\min} \min\{2x_{\min}, 1\} - \min\{2x_{\min}, 1\}^2\}$$

$$= \begin{cases} 5x_{\min}^2 & \text{if } x_{\min} < 1/2, \\ x_{\min}^2 + 4x_{\min} - 1 & \text{if } x_{\min} \geq 1/2. \end{cases}$$

On the other hand, we have

$$g_{\mathcal{Y}}(x) = \begin{cases} x^2 + 4x - 1 & \text{if } 1 < 2x, \\ x^2 + 8x^2 - 4x^2 = 5x^2 & \text{otherwise.} \end{cases}$$

For $x_{\min} < 1/2$, we find

$$\begin{aligned}\min_{x \in \mathcal{X}} \max_{y \in \mathcal{Y}} f(x, y) &= \min \left\{ \min_{x \in [x_{\min}, 1/2]} \{5x^2\}, \min_{x \in [1/2, 2]} \{x^2 + 4x - 1\} \right\} \\ &= \min\{5x_{\min}^2, 5/4\} = 5x_{\min}^2.\end{aligned}$$

If $x_{\min} \geq 1/2$, $g_{\mathcal{Y}}(x) = x^2 + 4x - 1$, and $\min_{x \in \mathcal{X}} \max_{y \in \mathcal{Y}} f(x, y) = x_{\min}^2 + 4x_{\min} - 1$, which coincides with $\max_{y \in \mathcal{Y}} \min_{x \in \mathcal{X}} f(x, y)$.